ALTO — Intelligent Real-Time Financial Decision Platform

PITCH DECK & EXECUTIVE SYSTEM ARCHITECTURE SPECIFICATION

Competition Track: Real-Time AI for Critical Decisions • **Target Architecture:** IBM Z / LinuxONE

■ **Live Production Web Application:** https://finance-g9aj.onrender.com	■ **Open-Source GitHub Repository:** https://github.com/haricharananubolu/Finance
■ **Evaluation Account:** demo@alto.app ■ **Password:** alto1234	■ **Live Status:** Operational • 19/19 Automated Pytest Tests Passing • Zero Third-Party Tracking

1. Problem Statement & Critical Decision Context

The Core Problem: Consumers handle thousands of rapid micro-transactions yearly through instant digital channels (UPI, contactless cards, digital wallets). Traditional banking platforms only deliver passive, retrospective statements 30 days after transactions take place. By that time, spending limits have already been breached and potential savings are permanently eroded.

Why Existing Tools Fail: Commercial budgeting applications monetize user data by harvesting transactions and injecting predatory loan advertisements. Users are forced to choose between invasive commercial tracking or cumbersome spreadsheets with zero real-time guardrails.

The ALTO Solution: An autonomous, privacy-preserving financial decision platform. ALTO intercepts transactions, recalculates all-time liquidity and period savings velocity in real time, and triggers preemptive multi-stage threshold warnings (Healthy <80%, Caution 80-100%, and Exceeded >100%) to halt budget breaches before critical financial damage occurs.

2. Target Beneficiaries & Measurable Value Delivery

College Students & Young Adults Managing tight monthly allowances/stipends. ALTO eliminates end-of-month cash crises through early 80% caution warnings.	**Early-Career Professionals** Juggling multi-channel expenses (rent, dining, EMIs). ALTO tracks precise savings rate percentages and 6-month comparative cash trends.
Privacy-Conscious Individuals Users demanding complete autonomy over financial data. Enforces confidential data isolation with zero commercial ad trackers.	**Enterprise Financial Organizations** Banks and fintech institutions seeking low-latency fraud and budget scoring engines deployable on secure IBM LinuxONE mainframe infrastructure.

3. System Architecture & Technical Implementation

Component Layer	Technology	Architectural Function & Guarantees
Ingestion & Web Layer	Python 3.12, Flask 3.0, Gunicorn WSGI	Application Factory with modular Blueprints (auth, dashboard, transactions, budgets, reports). Zero memory leak request dispatching.
Real-Time Decision Core	SQLAlchemy 2.0, Decimal Math Engine	Fixed-point Decimal math engine eliminates floating-point rounding errors. Instant evaluation of net balance & spending ceilings.
Persistence & Schema	SQLite / Relational ACID Database	Normalized relational tables with foreign keys and cascade delete constraints for Users, Transactions, and Budgets.
Security & Authentication	Werkzeug Scrypt, Flask-WTF, CSRF	Hardware-resilient password hashing, cryptographic CSRF tokens on all POST endpoints, and strict per-user query isolation.
Client Visualization	HTML5, CSS3, Vanilla JS, Chart.js	Interactive 6-month cash flow bar charts, category distribution doughnut charts, and responsive mobile navigation drawer.
Quality Assurance	Pytest 8.2 (19 Automated Tests)	100% passing test suite verifying login, CSRF tokens, threshold calculations, and cross-user data isolation.

4. How ALTO Leverages IBM Z & LinuxONE Technologies

1. Sub-Millisecond Real-Time Scoring (IBM Telum On-Chip AI)	For millions of concurrent digital payments, scoring spending against budget thresholds must happen in *sub-milliseconds* during authorization. LinuxONE's integrated on-chip AI acceleration enables ALTO to run inference directly alongside transaction workloads without offloading latency.
2. Confidential Computing & Pervasive Encryption	Financial records demand ironclad data protection. By utilizing IBM Secure Execution for Linux and CPACF (CP Assist for Cryptographic Function), ALTO isolates client financial logs in-memory and at-rest, providing cryptographic guarantees that even host administrators cannot inspect private financial data.
3. Enterprise Linux Container Scale (s390x Architecture)	Packaged as standard OCI container images running on Red Hat Enterprise Linux / Ubuntu for IBM Z, ALTO achieves 99.999% mainframe reliability, handling exponential transaction traffic with massive horizontal and vertical scalability.

5. Core Financial Formulas & Guardrail Engine

- **Dynamic Liquidity Balance:** Balance = ∑(Incomes) – ∑(Expenses) across all recorded timestamps.
- **Savings Velocity Index:** Savings Rate = ((Month Inflow - Month Outflow) / Month Inflow) × 100 (when Inflow > 0).
- **Autonomous Budget Threshold Triad:**
  - **Healthy Band (< 80% usage):** Normal spending trajectory with positive status badges.
  - **Caution Band (80% – 100% usage):** Instant amber alert prompting conscious spending adjustments.
  - **Exceeded Band (> 100% usage):** High-priority rose alert banner signaling an active financial deficit.

Production Deployment URL	https://finance-g9aj.onrender.com
GitHub Source Repository	https://github.com/haricharananubolu/Finance
Automated Test Suite Status	**19 Passed / 0 Failed** (Unit, Integration, CSRF & Isolation Tests)
Evaluator Login Credentials	Email: **demo@alto.app** • Password: **alto1234**